Bags, Vectors & Transformers

Introduction · Day 1, Part 1

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Today's Session

Welcome & Introductions

Who we are — and who you are

What is Computational Text Analysis?

The big picture

Why Should You Care?

Research examples and possibilities

What Could This Mean for Your Research?

Your data, your questions

Course Overview

The 4-day journey ahead

Practicalities

Tools, schedule, and expectations

Meet Your Instructor



Denise J. Roth
PhD Candidate · Strategic Communication Group
Wageningen University & Research

- Online political communication — incivility, polarization, public discourse
- Black digital health communication and scientization of public debate
- Computational text analysis as a method across all of the above

Feel free to ask questions throughout — this is a participatory course!

Meet Summer!

SS

Summer Skillen

PhD Candidate · Strategic Communication Group (COM)
Wageningen University & Research

Teaching assistant for this course

Error messages, Colab acting up, a notebook that won't run? Summer is there to help with the technical side during the sessions.

Now You!

Take a moment to introduce yourself to the group:

- Your name and department
- What you are researching
- What kind of text data you work with (or wish you could)
- Prior experience with methods or software (R, Python, SPSS… anything!)
- One thing you hope to get out of this course

What is Computational Text Analysis?

Using computers to systematically extract patterns and meaning from large collections of text — at a scale no human reader could manage alone.

What kind of questions?

· How does framing of topic X change over time?

· What themes dominate in corpus Y?

· How similar are these two sets of documents?

· Does sentiment differ across groups?

What kinds of text?

· News articles, social media posts

· Parliamentary speeches, policy documents

· Interview transcripts, open survey responses

· Pretty much anything in written form

The method is flexible — what matters is having a research question text can answer.

Quantitative or Qualitative?

Spoiler: it's both — and that's what makes it powerful.

The execution is quantitative

· Counts, frequencies, scores

· Algorithms and models

· Large N, reproducible

· Systematic and transparent

The interpretation is qualitative

· You still assign meaning

· You validate by reading

· Context always matters

· Numbers don't speak for themselves

Meaning-making is always qualitative

· Counting words $\neq$ understanding text

· CTA surfaces patterns

· You interpret what they mean

· The researcher is never out of the loop

Think of CTA as a tool, not a paradigm — it serves your research question, whatever that is.

Why Should You Care?

Analyzing 100,000 news articles to track how migration framing shifts over time

Comparing parliamentary speeches across 20 years and 5 countries

Classifying sentiment in millions of tweets after a political crisis

Making sense of thousands of open-ended survey responses without losing the nuance

What Could This Mean for Your Research?

Think about your own work for a moment...

- What text data do you already have, or could realistically collect?
- What research question could you explore if you could read 10,000 texts?
- Where does analysis currently take the most time — and could it be automated?

We'll come back to your answers throughout the week!

The Journey Ahead: 4 Days

DAY 1 · BAGS

Text as Data

- Python & Colab
- Bag-of-words & TF-IDF
- Where counting falls short

DAY 2 · VECTORS

Word Embeddings

- From counting to meaning
- Word2Vec & GloVe
- Measuring bias

DAY 3 · TRANSFORMERS

Contextual Embeddings

- Attention & BERT
- Hugging Face
- Fine-tuning a classifier

DAY 4 · AND BEYOND

Documents & LLMs

- Sentence transformers
- LLMs as annotators
- Ethics & open models

The Week at a Glance

MON Oct 12	**Bags of words** Introduction · Python & Colab · classic text analysis · bag-of-words and its limits	B0077
TUE Oct 13	**Vectors: static word embeddings** From counting to meaning · Word2Vec & GloVe · embeddings in social science · bias	**B0076**
WED Oct 14	**Transformers: contextual embeddings** Attention & BERT · transformers in practice · fine-tuning BERT on policy texts	B0077
THU Oct 15	**Sentence transformers & large language models** Document clustering with guest Prof. Art Dewulf · LLMs as annotators · ethics & open models	B0077
FRI Oct 16	**Optional: one-on-one sessions** 15-minute slots to talk through your own project, data or code (sign up in advance)	Online

Mon–Thu, 10:00–16:00 (lunch 12:30–13:30)

LeBo canteen: tell them you're with this course

Practicalities



Schedule

10:00 – 12:30	Morning
12:30 – 13:30	Lunch
13:30 – 16:00	Afternoon



Tools

Google Colab (browser-based, no installation needed!)

NLTK, scikit-learn, gensim, transformers

Links to notebooks provided



Prerequisites

Basic Python (DataCamp level)

No prior CTA knowledge needed — we start from scratch

Addressing the Elephant in the Room

Yes, you can use AI tools during this course. Most people who write code do.

It's how coding works now

- Coders have always borrowed: Stack Overflow, manuals, a colleague's script
- ChatGPT, Claude, Copilot and Colab's assistant are the newest version
- Great for explaining error messages

But it's still your analysis

- Code that runs isn't always right
- The worst bugs don't crash. They quietly give you wrong numbers
- In a paper, you have to explain every step, whoever (or whatever) wrote it

Some suggestions for this week

Try the exercises yourself first, then ask

Ask it to *explain* code, not only to write it

Keep sensitive data out of online chatbots

A simple test: if you can't explain what a line does, it isn't ready for your research yet.

Let's get started! 

Any questions before we dive into Python and text data?